A DIFFERENCE OF PULLBACKS IN AN INVERSE LIMIT OF SPECTRAL SPACES IS NONEMPTY EXACTLY WHEN ALL ITS FINITE-STAGE COUNTERPARTS ARE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-topology/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the Stacks Project keeps one single shared preamble for every chapter; this card takes it directly from the staged sources/stacks-topology/preamble.tex as three verbatim spans (lines 8-12, 14-18 and 154-217), so no derived file was written and nothing was added or redefined. The wrapper class and the base packages are supplied by the assembler.

Editorial note. Bibliography provenance: neither the counted statement nor its proof contains a citation (this lane rescans the whole span before writing the card), so the card carries no bibliography and no entry was reconstructed.

Editorial note. Selection provenance: this card was opened under the file-based control contract (CONTROL.md), with the mechanical pre-card gate run on its statement and proof interval before the card was written.

Allowed standard prerequisites (editorial).

- The statement that a set built as an inverse limit of nonempty closed sets in the constructible topology is nonempty. Invoked at the end of the counted proof (source lines 5086-5087). Converts nonemptiness of all the stages into nonemptiness of the limit.

Source reference. Original lemma lemma-inverse-limit-spectral-spaces-nonempty: sources/stacks-topology/pinned-original-source.tex, source lines 5027–5027. Navigation pointer only: the counted text refers to the Stacks item lemma-inverse-limit-spectral-spaces-nonempty, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $\mathcal{I}$ be a cofiltered category. Let $i \mapsto X_i$ be a diagram of spectral spaces such that for $a : j \to i$ in $\mathcal{I}$ the corresponding map $f_a : X_j \to X_i$ is spectral. Let $X = \lim X_i$ with projections $p_i : X \to X_i$. Let $i \in \mathrm{Ob}(\mathcal{I})$ and let $E, F \subset X_i$ be subsets with E closed in the constructible topology and F open in the constructible topology. Then $p_i^{-1}(E) \subset p_i^{-1}(F)$ if and only if there is a morphism $a : j \to i$ in $\mathcal{I}$ such that $f_a^{-1}(E) \subset f_a^{-1}(F)$.*

Proof. Observe that

$$p_i^{-1}(E) \setminus p_i^{-1}(F) = \lim_{a:j\to i} f_a^{-1}(E) \setminus f_a^{-1}(F)$$

Since f_a is a spectral map, it is continuous in the constructible topology hence the set $f_a^{-1}(E) \setminus f_a^{-1}(F)$ is closed in the constructible topology. Hence Lemma lemma-inverse-limit-spectral-spaces-nonempty applies to show that the LHS is nonempty if and only if each of the spaces of the RHS is nonempty. □

Date: stacks-project:topology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.